

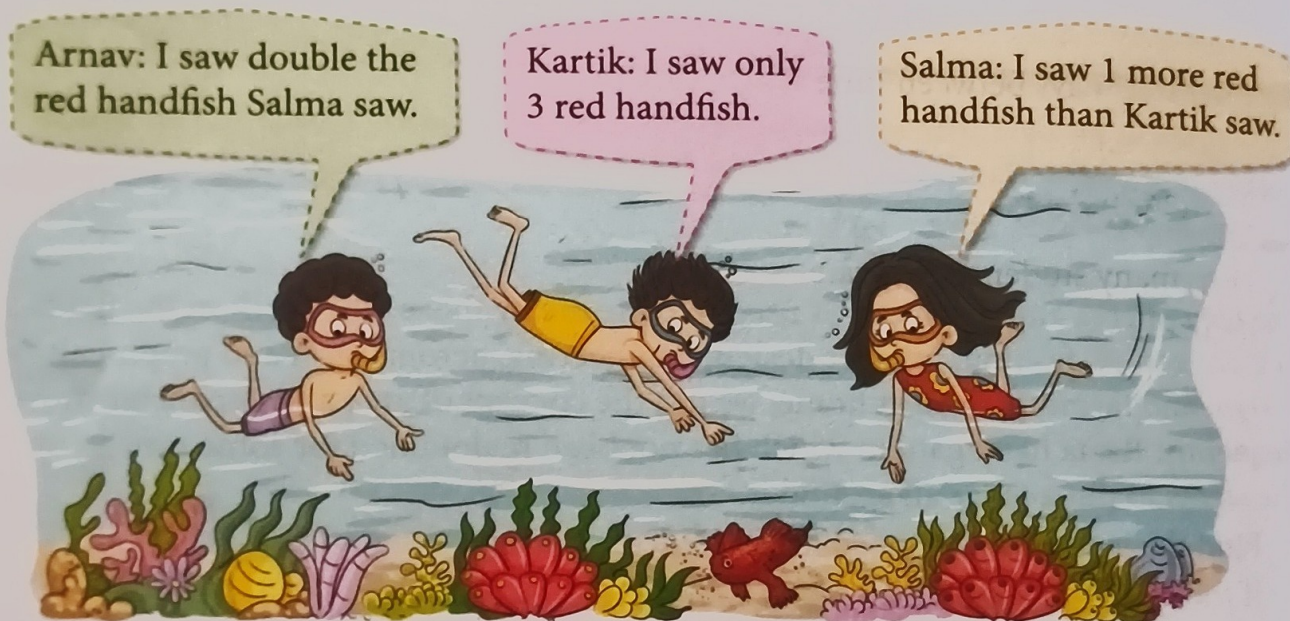
15

Data Handling

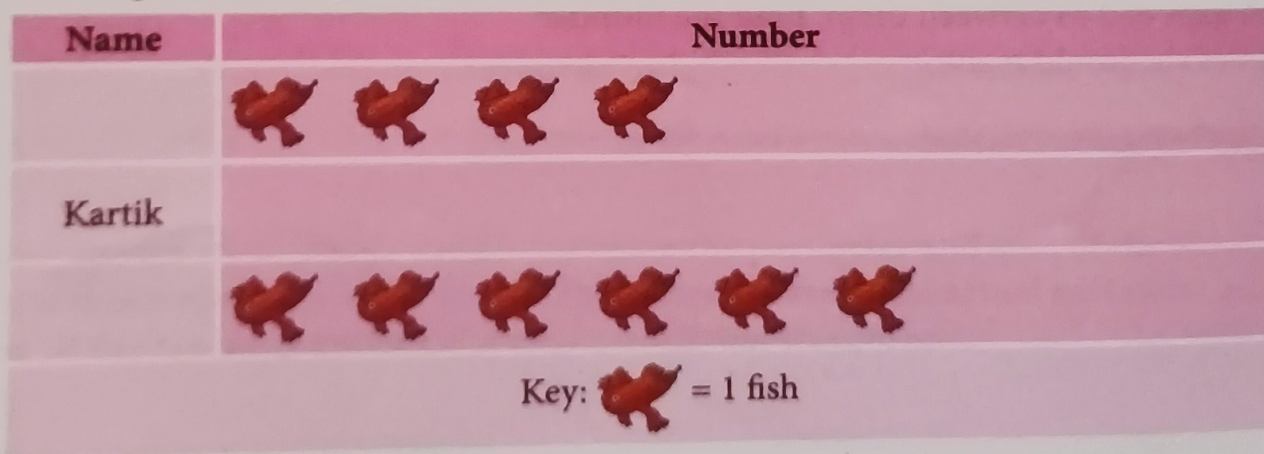
Tune In



The red handfish is one of the endangered species, which means that it is in danger of disappearing forever. Overfishing is the main threat to red handfish populations. Arnav and his friends went scuba diving with the diving instructor to discover the underwater world, which was an incredible experience.



Complete the pictograph below by writing the children's missing names and drawing the missing fish.



BAR GRAPHS

A diagram that uses vertical or horizontal bars of equal width to show the number of items under each category is called a **bar diagram**, also known as a **bar chart** or **bar graph**. The title explains what the bar diagram shows and the **labels** explain what the bars represent.

Vertical Bar Graph

The vertical bar chart below represents the number of students of Class 4A and Class 4B who have their birthdays between June and December.

The title of this bar graph is 'Birthday Chart of Class 4A and 4B between June and December' and the labels are June, July, August, September, October, November, and December.

Let us study the bar graph and answer the following questions.

1. How many students celebrate their birthdays between June and December?

Ans: 36 students

2. How many students celebrate their birthdays in July?

Ans: 5 students

3. In which months do an equal number of students celebrate their birthdays?

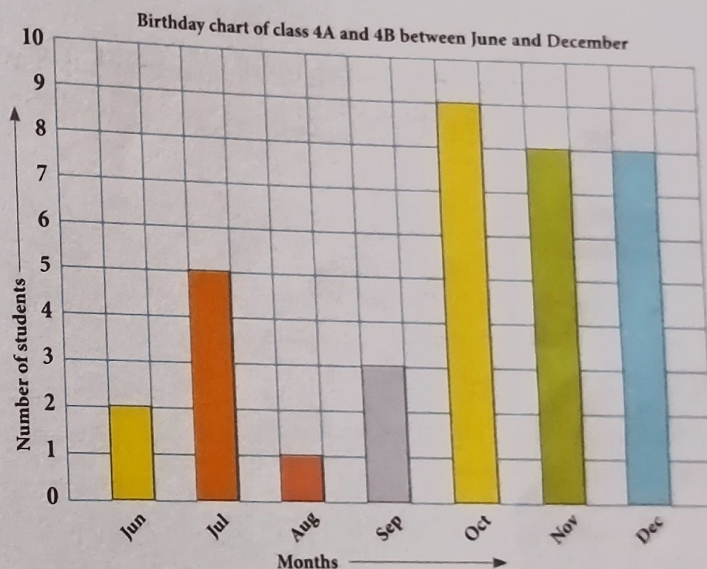
Ans: November and December

4. Which month has the least number of birthdays? How many?

Ans: August. 1 birthday.

5. Which month has the maximum number of birthdays and how many?

Ans: October. 9 birthdays.



Horizontal Bar Graph

Look at the horizontal bar diagram given below. It provides information about the favourite snack of a group of students. We can find out the number of students who like a particular snack by looking at the lengths of the bars.

Let's Try! 1

Shruti, Sanjay, and Shankar collected data regarding the most watched TV programmes in their neighbourhood. They prepared a bar graph based on the data.

Study the bar graph and answer the following questions.

1. Give the bar graph a title.
2. How many people were interviewed?
3. Which programme has the maximum viewership?
4. How many people watch Science programme and Cartoon show?
5. How many people watch News on TV?
6. What is the viewership for Breakfast show?



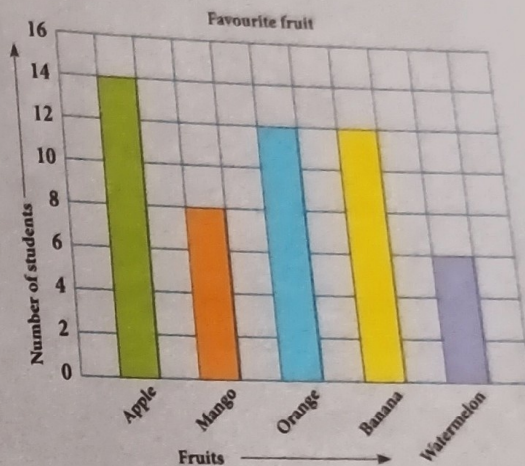
Exercise 15A

Go to MyInspire and attempt question 1.

A group of students was asked about the fruit they liked the most. The following bar diagram shows the results.

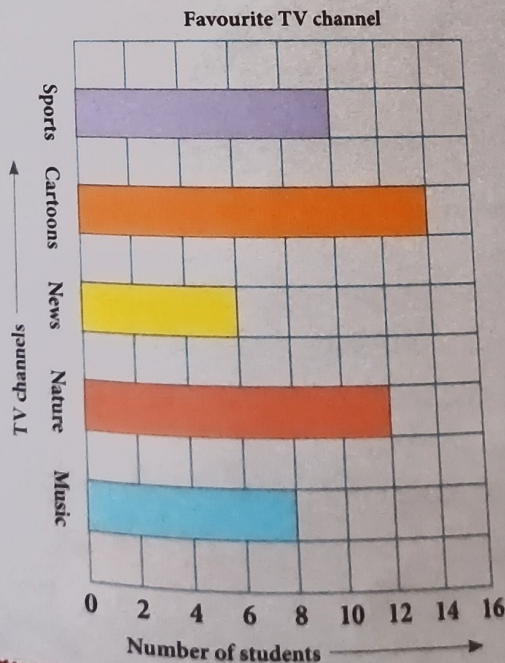
Study the bar graph and answer the questions.

- How many students were asked about their favourite fruit?
- Which is the most popular fruit? How many students like it?
- Which is the least popular fruit? How many students like it?
- An equal number of students prefer two fruits. What fruits are they and how many students prefer each of them?
- The number of students who like mango is two less than the number of students who like banana. State true or false.



- The students of Class 4A and Class 4B drew a bar diagram to show their favourite TV channels.

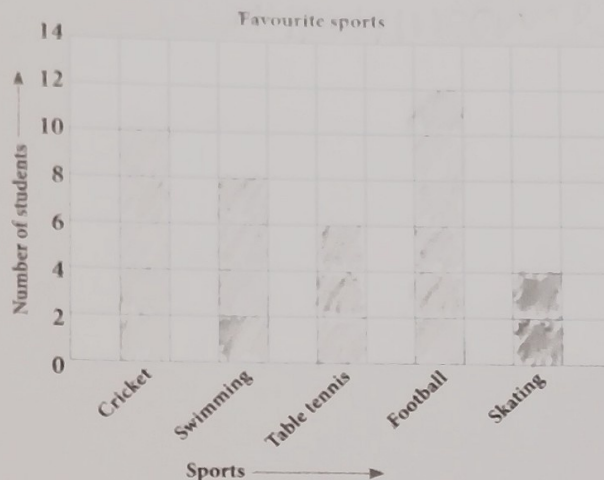
Study the bar graph and answer the questions.



- What is the strength of the class if five students were absent on that day?
- Which channel is liked by the most number of students? How many students like it?
- Which channel is liked by the least number of students? How many students like it?
- Which channel is liked by a greater number of students—Sports or Nature?
- How many students like music channel?

Application Based Questions

- The data about the favourite sports of the students of Class 4 is given below.
Cricket: 10 Swimming: 8 Table tennis: 6 Football: 12 Skating: 4
Draw a vertical bar graph in the given space.



5. A company organised a New Year sale from 27th December to 5th January. Details of the sales are given below.

Television set: 40

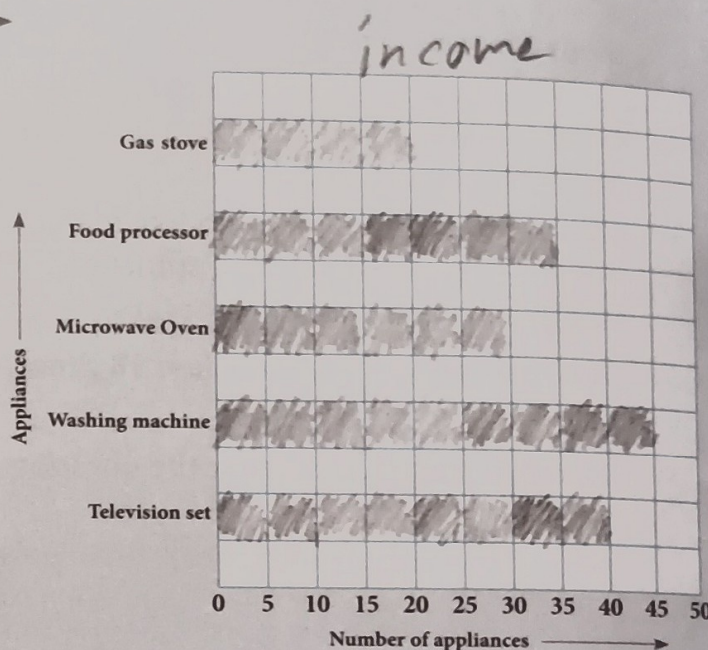
Washing machine: 45

Microwave oven: 30

Food processor: 35

Gas stove: 20

Draw a horizontal bar graph in the given space.



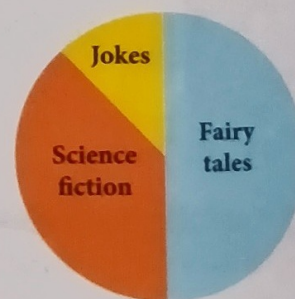
CIRCLE GRAPHS

A circle graph is a circular chart in which the circle is divided into parts. Like pictographs and bar graphs, circle graphs are also used to show and compare information. Circle graph is also called a **circle chart** or a **pie chart**.

Let us look at the given circle graph.

This circle graph shows the favourite books of students of Class 4.

- The blue part is the largest part of the circle. It shows that most of the students like to read fairy tales.
- The yellow part is the smallest, showing that the least number of students like to read jokes.
- The orange part in the circle graph shows that science fiction books are the second most popular among the students.



Example: The number of students in Class 4 who use a different mode of coming to school is shown in the circle graph.

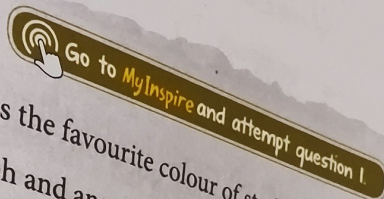
a. How do most students come to school?
Ans: Most students come to school by school bus.

b. What is the least popular mode of transport?
Ans: Least popular mode of transport is cycle.

c. The number of students who walk to school is equal to the number of students who come to school by car. State true or false.
Ans: True



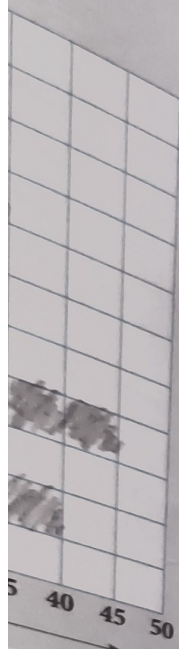
Exercise 15B



2. The circle graph shows the favourite colour of students in a class. Look at the circle graph and answer the questions.
- Which colour is more popular—blue or yellow?
 - Which is the favourite colour of most students?
 - Which colour do the least number of students like?
 - Which colour is less popular—green or blue?

Application Based Questions

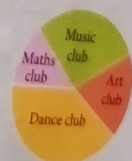
3. The circle graph shows the number of children in different clubs. Look at the circle graph and state true or false.
- Dance club has the most number of students.
 - Music club has the least number of students.
 - Maths club has the same number of students as Music club.
 - Music club has more students than the Art club.
 - Dance club has half of the students.



ictographs



second



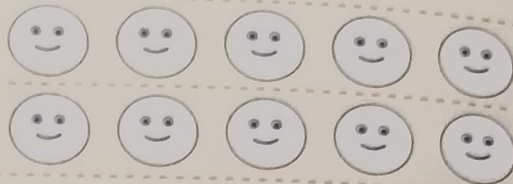
Let's Review



Come here after you have completed the chapter. Colour 1 to 5 😊 to show how well you understand a topic. (5 😊 = Very well; 1 😊 = Not at all)

Bar Graphs

Circle Graphs



Revision Exercises



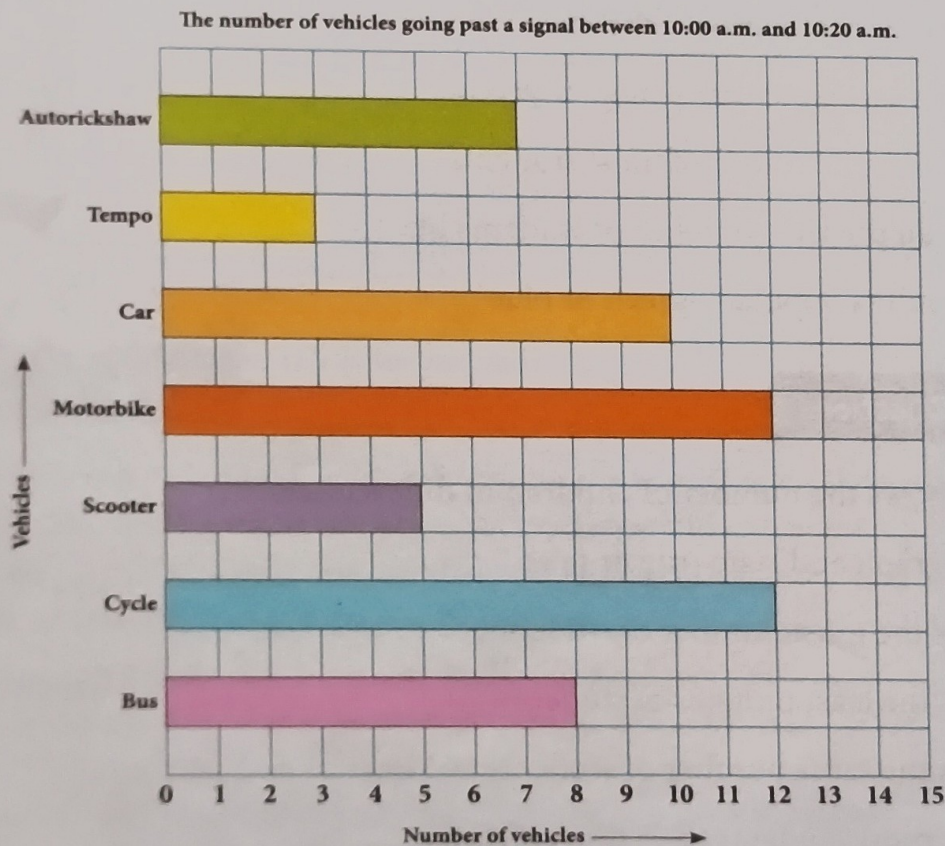
Go to *MyInspire* for



Start Up



1. Study the bar diagram below and answer the questions that follow.



- a. The number of scooters that went past in the 20 minutes = _____
- b. How many buses went past in the 20 minutes?

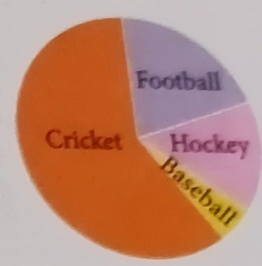
- c. How many scooters crossed the signal?
- d. What is the total number of cycles and tempos?
- e. The total number of vehicles that went past = _____.

Define the following terms.

- a. Data
- b. Circle graph

The circle graph shows the favourite sports of the students of Class 4.
Look at the circle graph and fill in the blanks.

- a. The most number of students like _____.
- b. The second most favourite sports is _____.
- c. The least number of students like _____.
- d. More than _____ of the students like cricket.

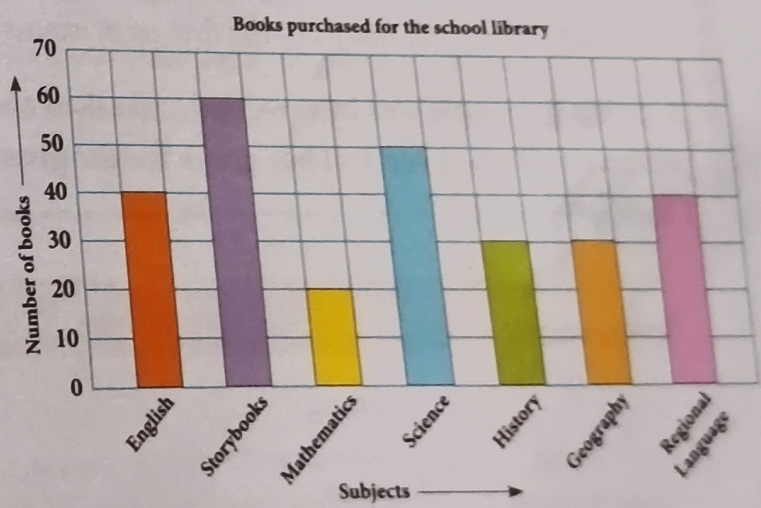


Level Up



1. Books were bought for the school library for the new academic year. The details of the purchase are given as a bar graph. Study the bar graph and answer the questions.

- a. How many books were purchased in all?
- b. How many storybooks were bought?
- c. The same number of books were bought in two pairs of categories. What are they?
- d. Which category got the least number of books?



2. During summer vacation, a school offers 4 different activities. The circle graph shows the number of children in each activity. Look at the circle graph and answer the following questions.

- a. Which activity has most number of students?
- b. Which activity has the least number of students?
- c. Which activity is more popular—dance or music?

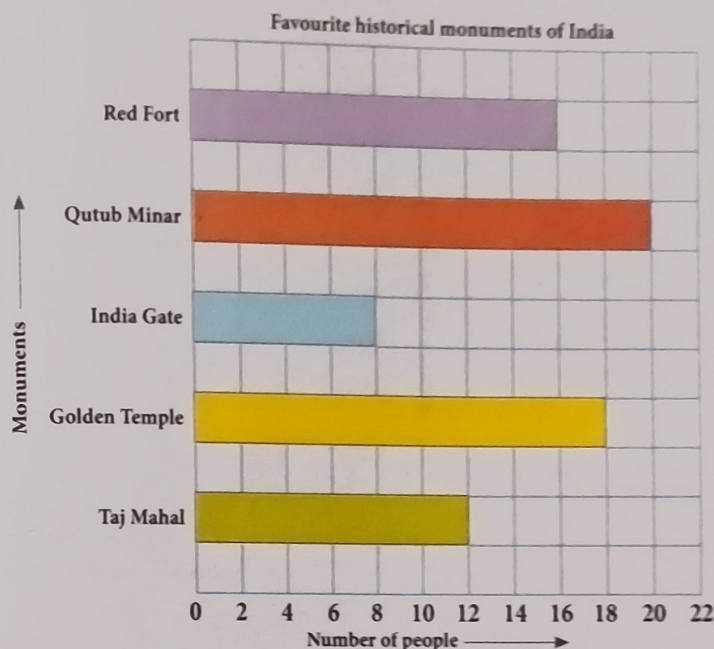


Exercise	Number of people
Walking	55
Jogging	20
Yoga	35
Aerobics	10
None	30

HOTS Questions



India is rich in culture and diversity. The country is home to some of the most beautiful historical monuments in the world. Nita surveyed favourite historical monuments in ABC school and collected the data in the form of a bar graph, as shown below. Study the bar diagram and answer the questions that follow.



1. If 17 students were absent on that day, calculate the strength of the school.
2. How will you represent the number of people on a bar graph if the number of people shown for each bar increase ten times?

Everyday Maths



Pictographs and bar graphs can be commonly found in newspapers and news reports. A graph is a very effective visual aid that helps us understand a situation. Graphs help us compare information, estimate, and predict a pattern to handle future situations.

Search a few newspapers and find a news article where a bar graph is used to represent a study. Cut out the bar graph picture and data. Paste them in your notebook. You can explain the article you had read to the class using the bar graph.

(EVS) Do you know that every child has the right to free education up to Class 8? Do you think that all children are able to study up to Class 8? Discuss the importance of education in the class.

The given bar graph shows how much Rohan's parents spent on various school-related items in a year.

Look at the bar graph and answer the following questions.

1. What is Rohan's monthly fees?
2. How much is spent on uniform?
3. Find the total amount spent on various school-related items in a year.

